

A Case Study on H2O OpenCasebook : Uncovering Digital Reading in Law School

Seonghee Lee
sl994@cornell.edu
Cornell University
Ithaca, New York, USA

Catherine Brobston
cbrobston@law.harvard.edu
Harvard Library Innovation Lab
Cambridge, Massachusetts, USA

Jack Cushman
jcushman@law.harvard.edu
Harvard Library Innovation Lab
Cambridge, Massachusetts, USA

Harmony Eidolon
heidolon@law.harvard.edu
Harvard Library Innovation Lab
Cambridge, Massachusetts, USA

ABSTRACT

This study investigates the digital reading experiences of law students and conducts a usability test on H2O, a digital open-casebook platform developed by the Harvard Library Innovation Lab. 9 in-depth user interviews and 16 user surveys on digital reading experiences of law students were conducted. The results of this study provide insights into the reading preferences and habits of law students that can be used to inform the design of digital learning tools in law school. Despite anecdotal reports that earlier users of the platform preferred physical casebooks, we found that many law students now prefer using digital casebooks. More than half of the participants (53.3%) in our study reported that they preferred digital casebooks or that the format did not matter. Even for those preferring print books, cost or weight reduction was often more important than format preference, and print-preferring users expressed an interest in using digital annotation tools, collaboration and additional search features. This study suggests that student preferences for learning materials are rapidly changing — a change that may have been accelerated by the pandemic — and that assumptions about student preference may need to be reevaluated. It also demonstrates the importance of comparing old vs. new offerings, such as print vs. digital, to evaluate the strength of the preference relative to other preferences such as cost, and to teach participants what new features are enabled by the new format.

CCS CONCEPTS

• **Human-centered computing** → **Natural language interfaces; Empirical studies in interaction design; Interaction design process and methods.**

KEYWORDS

digital reading, law school, learning technology, digital reading interaction

ACM Reference Format:

Seonghee Lee, Jack Cushman, Catherine Brobston, and Harmony Eidolon. 2023. A Case Study on H2O OpenCasebook : Uncovering Digital Reading in Law School. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3544549.3573857>

1 INTRODUCTION

Casebooks in law school are a significant financial burden to law students. Most legal coursework is taught from casebooks that cost around 200-300 dollars each, despite being comprised of case laws that are already in the public domain. Additionally, many of these casebooks are not perfectly suited for the classroom — often professors will not use the entirety of the casebook but resequence and selectively assign readings from the book. Despite this, many law school classes require students to purchase physical casebooks, which means that they will be carrying heavy weights of books on a daily basis to class.

H2O Open Casebook attempts to solve this issue. H2O is a free platform for making, sharing, and remixing open access and open-licensed casebooks and other course materials developed and maintained at the Library Innovation Lab at the Harvard Law School Library (Figure 1). It provides law professors with numerous cases from CAP (the Caselaw Access Project), and allows them to select cases themselves, add personal notations, and publish these books online for free. Students can then access these casebooks online and read them digitally or export a Word copy for printing. By creating and sharing casebooks and other resources in H2O, educators can drastically reduce textbook costs for their students, collaborate toward new pedagogical methods, and increase the impact of their ideas.

One issue that the Harvard Library Innovation Lab had been facing was in understanding how law students were reading H2O casebooks. Law students spend many hours per day reading during their time in law school, so are sensitive to the specifics of book formatting, and in prior years the project has received anecdotal feedback that law students prefer physical books to digital books.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '23, April 23–28, 2023, Hamburg, Germany

© 2023 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3573857>

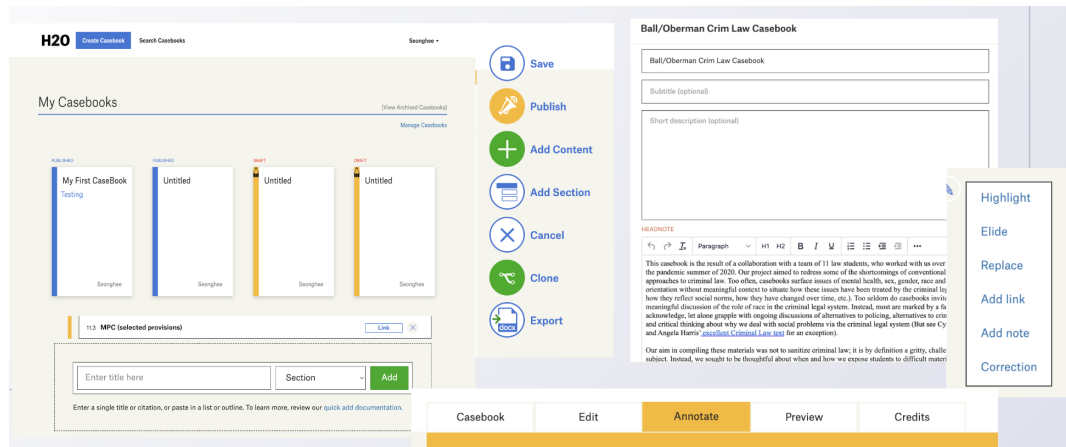


Figure 1: The Digital Interface of H2O Open Casebook. Law Professors can add specific cases, edit, and add contents to their book. A published H2O Casebook is available on the website for the public, and can be exported to a paper copy.

However, the widespread use of digital devices and the shift to remote classes raised the need to test this received wisdom. Additionally, the format in which law students experience their casebooks can provide different affordances for law students as they do their digital reading, such as annotation, search, and collaboration with others. To understand casebook format preferences and to understand the digital reading interactions of law students, we conducted a H2O usability test and survey.

In order for digital casebooks to become successful learning tools, they need to take the unique active reading process of law students into consideration. The results of this study contribute to the body of research on digital reading of law students, a unique group of readers with a focus on heavy amounts of active reading. Understanding their needs will inform design research on how we can design digital learning tools for the law school classroom.

2 BACKGROUND

Law schools require a great amount of active reading. Law students are expected to self-learn skills such as factual understanding, case briefing, legal reasoning, and drafting, using active reading techniques such as highlighting and summarizing [3][2]. The use of technology has been controversial in the law school classroom [12]. Those skeptical of the use of technology argue that students tend to fixate on Powerpoint slides for memorization rather than making an effort to think and reflect critically on the material [11]. Additionally, in recent years, online resources such as Quimbee generate case briefs automatically, making it unnecessary for law students to go through the process of self-learning the elements of a case brief themselves [7].

Despite these concerns, the use of technology in the law school classroom has opened doors for many professors [9][5]. For example, professors are able to bring outside material into the classroom, such as spreadsheets and videos [10]. Additionally, anonymous

discussion threads such as Nymity have been used in the law classroom to encourage discussion on sensitive topics at Harvard Law School [4]. These digital tools are also beneficial for students[8]. Digital open-access casebooks have provided students with the opportunity to gain access to casebooks without having to pay expensive prices [1]. Additionally, digital learning tools have assisted law students in collaborative learning and in having portable, easy access casebooks [6].

Despite the increased use of technology in the law school classroom, there is a lack of research surrounding how we can design better learning tools for law students. Law schools have very specific teaching methods such as the Socratic Method and the skill sets law students have to develop are very specific for legal studies. More research is needed to look into how future digital learning tools could be used to support active reading processes of law students.

3 STUDY METHODOLOGY

Two studies were conducted to understand the student reading experiences on the H2O platform. An online survey was conducted to understand casebook format preferences and gain insight into the active reading annotation systems used by different law students. And a usability test evaluated student use of H2O features, the nuanced interactions students had with digital readers, and how each student's annotation system influenced their digital reading preferences (Figure 2). The objectives of this study were to learn the casebook format preferences of law students (digital vs. physical), active reading processes, annotation habits, and other factors that affect learning processes. The results informed the design of H2O itself as well as providing insights into the design of learning technology for the law school classroom.

3.1 Study Recruitment

The online survey and usability test participation form was sent to a range of law students that were currently enrolled or had recently graduated from law school. These recruitment emails were sent to

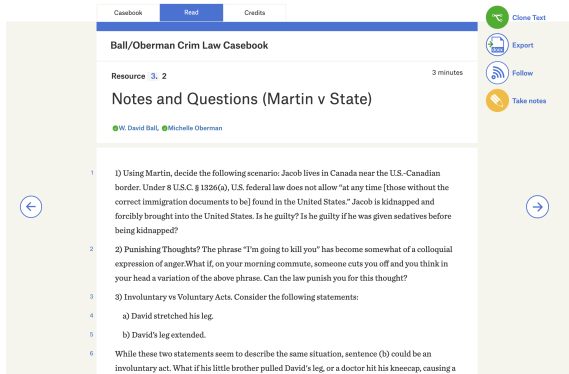


Figure 2: The Reading Interface on H2O. Students can annotate cases, print paper copies, and navigate through the book with page buttons.

personal email accounts gained from law school admission office websites and through admin contacts. We received participant responses from more than 10 different law schools including Berkeley, Harvard, Yale, Cornell, Duke, Emory, Penn, and Boston University. A total of 21 law students participated, 16 in the online survey and 9 in the usability study, with 4 overlapping participants.

3.2 Online Survey

The online survey asked questions about casebook format preferences, differences in information retention based on the casebook format, annotation habits, and digital e-reader features. The survey was sent to students at 10 different law schools and we received a total of 16 survey responses for this survey.

3.3 Usability Test

The second study was a usability test on H2O's platform. We interviewed a total of 9 law students from 5 different schools (Cornell, Boston, Duke, Penn, and Emory University) who were rising 2L/3L students or recent graduates. The study was conducted for a total of 45 minutes and participants received a 35 dollar Amazon gift card as a participation gift. We asked participants about their preferences in reading casebooks (digital v. physical) before the study. Participants were asked to perform a series of tasks on the H2O platform such as searching for specific content, annotating the casebook, and navigating through different content on the casebook. After interacting with H2O, we asked participants which format they would use H2O.

3.3.1 Information Search within the Casebook. Participants were asked to search for all the cases in the casebooks that had the word "train". H2O offers a search inside casebook feature that allows users to search for information inside the casebook. Search within specific materials such as legal documents, text passages, and links are also offered. Searching within a casebook is a useful feature for law students to be able to compare across cases and find similarities or trends.

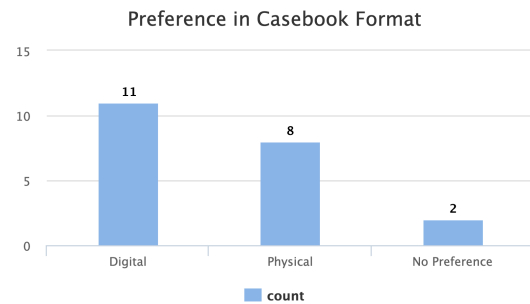


Figure 3: Preferences in Casebook Format

3.3.2 Page Navigation within the Casebook. One of the affordances of a physical casebook is that it allows for users to flip between sections quickly. The current user interface of H2O does not have a quick way for users to do this. By conducting the user study, we asked law students to navigate to different parts of H2O, to get an idea of how users would be navigating to different parts of the content and also an idea of how they preferred to navigate through casebooks.

3.3.3 Annotation of the Casebook. Making annotations on different cases is a crucial part of learning in law school. While H2O did offer some annotation features, most of these features were focused on professors and the annotations that would be useful for them to make in the process of publishing their casebooks. Therefore navigating to the part of H2O to make annotations required students to follow a workflow of a professor creating a casebook. Through the usability test, we aimed to get an idea of how the user expected to annotate, what types of annotations were needed and additional processes of annotation.

4 RESULTS

4.1 Casebook Format Preferences

Overall, we observed mixed responses in student preferences for casebook formats with slightly more students preferring digital to print (Figure 3). However, when other factors such as cost and weight were included, many students tended to prefer digital formats. Out of the total of 21 participants in both the survey and interview (4 participants participated in both), 11 participants preferred digital casebooks and 8 participants preferred physical casebooks. 2 participants showed no strong preference for either of the formats.

The main reasons behind a stronger preference for digital casebooks included lighter weight, easy transportation of text, efficient search, and accessibility. "I don't have to carry a heavy book. I can read it on my phone, laptop, tablet, wherever I am. I can make notes on the pdf, print, and also copy paste text from the readings." On the other hand, students that preferred physical casebooks reported such preference due to disability symptoms such as migraines. Other reasons included freedom of annotation (multiple highlights, personal markings) and ease of spatial navigation

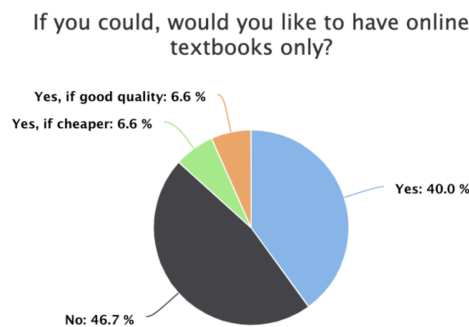


Figure 4: Preferences on Changing to Digital Casebooks

through the book. “I alternate between physical and digital because of the weight of physical books and my disability symptoms - migraines.” Some preferences were also attributed to familiarity having interacted with physical textbooks in past classes. Some students showed hesitation to transition into fully digital casebooks due to past experiences with difficult interfaces in digital casebooks.

Overall, 60% of students replied that they would like to move to digital casebooks (Figure 4). Students expressed dissatisfaction with having to carry heavy casebooks on a daily basis. “I would like to move to all e-books if it is possible. I do not like to use physical textbooks because of its weight.” “My back pain hates carrying around law school textbooks every day.” Other students expressed satisfaction with additional features offered by digital casebooks. “I really appreciate the short quizzes some services offer with the textbook, as well as short case briefs that include the analysis/tests the case uses.”

4.2 The Active Reading Process of Law Students

We observed a general pattern of an active reading processes that law students followed. Students started off by annotating sections of the casebook to mark main parts of the case such as the parties, facts, and holdings. Students would then add additional annotations of things mentioned in class and personal notes. There were a variety of different reported methods that students used to do this. While some students used detailed annotation processes with various highlights and personal markings, others used a simpler annotation method, leaving letter markings of what case details (evidence, holdings, parties) a certain part of the case referred to. Simple annotation systems were common among upperclassmen. Towards the end of the study, usually in preparation for exams, students created case briefs. Case briefs are summarizations of cases that distill the case’s most important parts. In preparation for these case briefs, main quotes of the casebooks were often copied so that students could refer back to the quotes. Preparing these case briefs was a crucial part of learning how to read case laws.

4.3 Formats for Annotation

We observed different preferences on using physical or digital casebooks for annotation. Some students preferred physical casebooks

for detailed annotation (multiple highlights, personal markings) while others preferred digital casebooks for the ease of search and transportation of text. All of the students in the usability test reported using a digital format to create their case briefs. Even students who preferred physical casebooks responded that they create case briefs in online editors. The main reason for this was the ease of organization and speed of digital editing systems. “I outline as I go through the course, so all notes go into my Word Doc outline. Specifically Word because I am a PC User and the headings and styles are crucial to my organization and navigating my notes. Also, most exams are taken on a computer with access to personal notes.” Additionally, the ability to copy text directly from the casebook to add quotes to the case briefs allowed students to save time and record more accurate information. “I can copy and paste parts of the case and have direct quotes in my case briefs that I can refer back to. It is more accurate and quicker to do that by taking notes digitally rather than physically.” With better understanding of how modern students use casebooks to prepare for class and for exams, digital textbooks could be designed not just as passive inputs to the active reading process, but as tools that help scaffold effective learning behavior.

4.4 Student Annotation Systems

First year law students tended to have more detailed annotation systems. One of our participants in our usability test was a rising second year law student just out of her first year of law school. She had a complex annotation system where she used up to 9 different highlight colors as well as personal markings to remember information. Due to the fact that a lot of her first year classes required her to have physical casebooks, she had only interacted with physical casebooks. The process she used to study was to first annotate her casebooks with her multiple highlights. Each of the highlighted colors pointed to different parts of the case’s main information such as the facts, holdings, and evidence. She would then leave side notes in the margins about topics discussed in class. When she studied she had her laptop computer at her side so that she could organize her case outlines in a separate OneNote document. She also mentioned the use of study groups in law school where she could share notes and ask questions.

While first year law students tended to have detailed annotation systems, we observed a trend of less complex annotation systems in upperclassmen. One of our participants who was a recent graduate of law school reported using no colors to highlight at all, but markings in the margins of casebooks to signal parts of the case such as the facts, holdings, and parties. This participant mentioned that the purpose of annotation was for easy retrieval of information from the text. Now that she was more accustomed to reading cases, she no longer needed complex annotation systems to help her learn to read legal cases. Her focus was now more on creating case outlines/briefs in a more efficient way. She also mentioned that she preferred digital casebooks because they were less stressful on her back, and that most of her classes were seminars, making casebooks less necessary. Additionally, the process of creating these case outlines often involved summarizing parts of the text or copying and pasting quotes from the text directly onto the case outline,

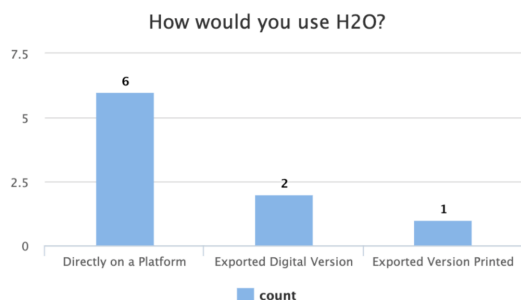


Figure 5: Interview Responses after Usability Study on how would you use H2O?

for which a digital casebook was more efficient. She also mentioned that she liked the feature in some of her digital casebooks that copied along the link to the section of the book or case from which the quote was copied so that she could refer back to the case later.

While the complexity in the annotation systems differed for law students in different years, all of our participants mainly used their annotations in the process of creating case briefs. And all the law students in our usability test used a digital format to make these case briefs due to the easy and quick organization that was possible through the digital platform, suggesting that improvements to the digital annotation experience would be particularly valuable.

4.5 How would you use H2O?

After completing the user study, we asked students about how they see themselves using H2O, and whether they would be most inclined to use H2O's digital or print options if assigned to read an H2O book. The results, surprisingly, did not align with their declared preferences for casebook format. Even participants that showed stronger preference for reading casebooks physically, after interacting with H2O, saw themselves using H2O digitally (Figure 5).

Most of the participants saw themselves reading casebooks both directly online through H2O and by exporting it. 4 participants responded saying that they would use H2O directly on the platform, 2 responded they would read H2O digitally through the exported version, and just one participant responded that they would print the casebook out to use it. 6 out of 9 participants responded that they saw themselves using H2O directly on the platform.

Surprisingly, all the participants who preferred to read casebooks physically responded that they would like to read H2O directly on the platform. The two main reasons for this were the personal notebook creation and collaboration features offered by H2O. "I really like the idea of cloning the case to your own notebook online and creating a document of it digitally." In H2O, students can clone parts of the casebook into their own notebook and create annotations from there (Figure 6). This allows students to create a separate notebook for themselves where they can create their own notes

while having the direct text with them. Students can additionally add, remove, and edit parts of the text to fit their learning purposes allowing for more freedom of customization within the book. Additionally, H2O allows students to collaborate with other students on this notebook. From here students can add other collaborators onto their book and share documents with each other. "I could see myself working on this notebook with other law students too".

When we asked these participants about why they preferred to read directly on H2O despite their preference for reading casebooks physically, they responded that access to casebooks was a greater need than having a preferred format and that the additional affordances offered by H2O offered greater benefits for them. 2 out of 9 participants responded that they would read H2O digitally through the exported version. These two participants preferred to read casebooks digitally and found the annotation features of H2O confusing. While many e-readers allowed for direct annotation on the text, H2O required users to go through a series of steps in order to annotate a casebook such as creating a separate notebook, cloning the case, and navigating through the annotation section. This process seemed confusing for the participants who were more used to an easier flow of annotation in other digital platforms. 1 out of 9 participants responded that they would use H2O through a printed exported version. This participant preferred to read casebooks digitally and found that the annotation features offered by H2O were not enough to fill her needs. "If there were a more robust set of highlighting tools and adding notes in the annotation, I would see myself using it. If I could highlight a section and add notes and if the notes could directly link back to highlights."

5 DISCUSSION

5.1 Design Considerations

5.1.1 Reading Patterns of Law Students. When designing digital casebooks for law students, we found that it was important to consider the reading patterns of law students. Law students often use their casebooks to initially read through, skim, and mark main points of the case (such as facts, evidence, parties, and holdings) in order to prepare a case brief. When law students are reading a casebook digitally, they are often able to highlight these points on a casebook and copy the sections into their digital casebook briefs. When parts of the text are copied, having a link back to the original source being automatically copied helps law students easily refer back to parts of the case. First year law students tend to focus on training themselves to read through cases effectively, and we often observed more detailed annotation systems used for first year law students. On the other hand, upperclassmen tended to focus more on identifying key information within the cases quickly and therefore their annotations were more focused on helping them identify key parts of the text more quickly and on creating case briefs easily. Finally, it is important to consider the use of case brief summarization tools available online in the learning process of law students, such as Quimbee. We often observed how these summarization tools were utilized in the study process of law students.

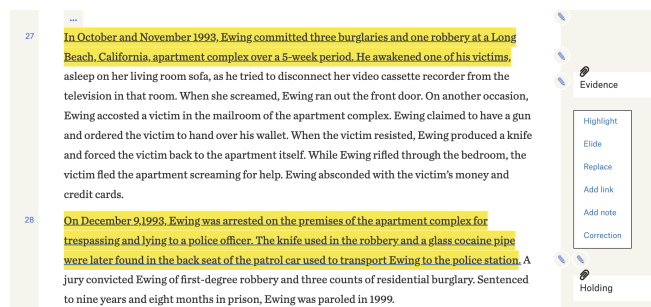


Figure 6: The Annotation Interface on H2O. Add notes, links, and highlight features offered.

5.1.2 Collaboration Features. Collaborating in the study process was also very common among law students. In preparation for exams and in the process of creating case briefs, law students often reported using online collaboration tools such as Google Docs. In the creation of effective learning tools for law students, thinking of effective ways for law students to collaborate in the learning process is important. We also observed how law students often had case outlines that were passed down from upperclassmen to lower-classmen. When preparing case briefs, this is where law students would often start making their annotations. Online learning tools could consider how to incorporate knowledge passed from earlier readers to later ones.

5.1.3 Law Students and Annotation. The most used annotation feature for law students were highlights. Participants in our usability study reported to using an average of 3-4 highlight colors. Since the purpose of annotating casebooks was often to make case briefs, the highlights were often used to emphasize main parts of the case for future reference. One pattern we observed in law students was that many of them had their own personal system of annotations. We observed different marking patterns, underlines, and letter markings used. However, when law students tried to use these annotations on digital casebooks, they were often limited to the few annotation features provided by the digital platform. Expanding the set of interaction tools for law students that can assist students in developing their own personalized annotation systems is key in helping them develop effective personal strategies for learning to read case laws. In future designs of digital casebooks, novel annotation methods that help law students systematically analyze cases to assist their learning may be useful in helping students learn effectively.

6 CONCLUSION

In this work, we aimed to provide insights into the usage of casebooks in law school and the digital reading habits of law students. The findings of this work provide insight for digital casebook design. We conducted 9 in-depth user interviews and 16 user surveys on digital reading experiences of law students. The survey asked current and recent graduate law students about their casebook format preferences, annotation habits, and digital reading experiences. During the usability test users were asked to do a series of 7 different tasks on the H2O platform such as digital annotation, page navigation, and information search. Our work highlights the study

process of law students and serve as a point of reference for design considerations for legal education technology. Our findings show that there was significant student support for digital casebooks and that access to casebooks was often a greater need than the format. We observed the convenience of digital casebooks in creating case briefs, law students' personalized annotation systems, and their collaborative learning processes. Based on our findings, we highlight the need for digital casebooks in law school to incorporate designs that assist case brief creation, personalization of annotation, and collaboration.

REFERENCES

- [1] Matthew Bodie. 2005. The Future of the Casebook: An Argument for an Open-Source Approach. *Journal of Legal Education* 57 (03 2005). <https://doi.org/10.2139/ssrn.691985>
- [2] Kate Cain, Jane Oakhill, and Peter Bryant. 2000. Investigating the causes of reading comprehension failure: The comprehension-age match design. *Reading and Writing* 12, 1/2 (2000), 31–40. <https://doi.org/10.1023/a:1008058319399>
- [3] Paul Caron and Rafael Gely. 2004. Taking Back the Law School Classroom: Using Technology to Foster Active Student Learning. *HeinOnline* 54 (05 2004).
- [4] Harvard Berkman Klein Center. 2021. Nymity. (2021). <http://www.jstor.org/stable/42898029>
- [5] Yvonne Dutton and Margaret Ryznar. 2019. Assessing online learning in law schools: Students say online classes deliver. (2019). <https://doi.org/10.31228/osf.io/bm8s5>
- [6] Jessica Gilgor. 2018. Technology for Studying: How to Use It to Your Advantage. *HeinOnline* 47 (05 2018).
- [7] Andres Gonzalez. 2018. Law School a Journey like No Other. *Student Lawyer* 47 (2018).
- [8] Helen Hargreaves, Sarah Robin, and Elizabeth Caldwell. 2022. Student perceptions of reading digital texts for university study. *Journal of Learning Development in Higher Education* 24 (Sep 2022). <https://doi.org/10.47408/jldhe.vi24.817>
- [9] Emma Jones, Francine Ryan, Hugh McFaul, David Vince, Lawrence Kizilkaya, and Jamie Daniels. 2022. Using virtual reality to enhance the law school curriculum. *Teaching Legal Education in the Digital Age* (2022), 26–35. <https://doi.org/10.4324/9780429351082-3>
- [10] Charles D. Kelso, Charles H. Kelso, and J. Clark Kelso. 1985. How Computers Will Invade Law School Classrooms. *Journal of Legal Education* 35, 4 (1985), 507–513. <http://www.jstor.org/stable/42898029>
- [11] Douglas L. Leslie. 2000. How Not to Teach Contracts, and Any Other Course Powerpoint, Laptops, and the Casefile Method Approaches to Teaching Contracts. *Saint Louis University* 44 (2000).
- [12] Michele Pistone. 2015. Law Schools and Technology: Where We Are and Where We Are Heading. *Journal of Legal Education* 64, 4 (2015), 586–604. <http://www.jstor.org/stable/24716714>